Painting from Part

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***Abstract* — The chosen paper analyses the matter of painting the complete image from a piece of it, this method being called *painting from part / part-painting*, using each inpainting and outpainting. The authors propose a unique part-painting solution, given the connection between *part* and *whole*. The method shows effectiveness and progressive performance on the problem of painting from part, even considering irregularly shaped components.**

Keywords — painting from part, part-painting, part and whole

I. SOLUTION DESCRIPTION

Part-painting is easily figured by the aptitude of people, who given a portion of a picture, can also visualize the rest of the image that cannot be seen. The method has the role to create a complete and real image, using only one part of it. Thus, it has various applications in computer science: graphic generation for video games, artificial intelligence, generated media, but also in the visual arts: for obtaining abstract art, comics, animated movies, cartoons, and books. Therefore, to apply the solution, it is needed to get particularities from a local domain (the part received as input) and a global domain (other related images), creating a convenient correspondence between both.

The part-painting solution designed by the authors has the ability to generate the complete picture from an irregular shaped component. In scientific terms, they use the given part, a mask and random noise from a standard normal distribution to produce a consistent, credible, and practical image.

The structure of the method includes an encryptor to derive the distribution and characteristics from the part, a drawing application to generate the complete picture from noise and discriminators for the whole and part to differentiate the features of the whole picture and the known portion. The process is split into three phases: part-noise restarting (the system is restarted in the beginning to adjust normally distributed part-noise), part-feature repainting (the process repaints the produced characteristics by feature transfer in various overlays) and part-patch refining (the application clarifies the resulted image by connecting the very much alike pieces).

A. Part-noise Restarting

To show Generative Adversarial Network's considerable capacity to represent and generate images, the authors begin with painting from a noise and not straight from the given part. Moreover, they re-proceed the painting process using the part-noise taking advantage of the distribution of the local domain (as in the given component). Then, they review the noise with specifical parameters used by the encryptor, utilizing a formula that includes the Hadamard product for matrices. Thus, the scientists construct distributions for each picture to improve the painting.

B. Part-feature Repainting

Using GAN's process, the application imitates the human's ability to complete a region of an image by adding unseen elements to it, in this way controlling better the aspects of the part. To do this, they invented a new component, called *Repainting Residual Block (RRB)*, which given the complete characteristic produced by the last block of painting application and the part characteristic obtained from the encryptor for the specific overlay, results into a repainted characteristic of the whole picture. In this way, a domino effect of the features occurs, keeping an equilibrium between all of them, simultaneously using transferring procedures, and concatenating, separating, and combining techniques. In order to make the part-feature transfer, the calculus uses mean and standard deviation of each transferred feature. The whole-feature reconstruction combines convolution and parameters connected to the feature.

C. Part-patch Refining

For this phase, they "recycle" the part-feature to make the painting clearer, following these steps: firstly, they adjust the whole-feature presented above, to turn the pieces into pixels; then, the obtained pixels are used to take place of the pixels from the repainted portion, this process being named *part-patch transmit*. The procedure explained above can encounter difficulties if the known and the unseen part differ a lot. That is why the authors combine the transmitted parts and the whole pieces repainted in the second phase, into the *whole-patch fusion* process. After all these steps, the application returns the complete image as a result.

Experiments made by the scientists have shown that, analyzing the inpainting and outpainting methods with their new technique, their study results into a more competent mode to generate complete images from only one part, as it can be applied even on irregular shaped elements and, also, it uses improved datasets. These experiments prove that their method obtains more intelligent, practical, and genuine whole pictures.

II. RELATED WORK

A. Inpainting and Outpainting

Current studies for picture inpainting and outpainting are principally branched in non-learning and learning techniques. Non-learning methods might have positive effect for constructing a large digital image from a small sample image by taking advantage of its structural content, but they are hard to generate considerable results only using details provided by the given part. Moreover, these solutions are not applicable for complicated or sophisticated pictures because of the fixed dataset.

For the inpainting method, the unseen part is sometimes little and placed within the portion, being ready to occupy a limited collection of absent pixels, by computing neighboring pixels that have compatibility with the ones that are nonexistent, thus obtaining a remarkable outcome.

On the other hand, for the outpainting method, the undiscovered portion is located externally to the part and sometimes massive, fact that makes the approach challenging. This practice uses two phases: lengthening the region by component enlargement and producing the effect with machine learning techniques that attempt to exploit models by working with obtainable model information.

In accordance with the details given above, both inpainting and outpainting solutions are difficult to be practiced to each other: the first one presented is problematic to obtain big considerable details externally, while the second method cannot deal with irregularly shaped portions, because the input part is necessary to be squared. Furthermore, the couple techniques evaluate the portion one time only at the start, this way not obtaining sufficient details from the input that could be beneficial for the final painting.

The article approaches painting from part essentially counting on the subsequent considerations:

1. All aspects of the component (lower-ranking or higher-ranking) are firmly and closely related to the elements of the complete picture.

   and

2. Little pieces of the part have a tremendous possibility of generously persisting in the whole image.

To solve the issues mentioned above, scientists all over the world have improved deep Convolutional Neural Networks to deal with the part-painting problems. Thus, learning techniques are more advanced and straightforward. For example, Yu exposed the idea to fill imperfections with practical components from the known portion; Liu and Yu developed a unique mode to build CNNs that have the capacity to complete irregular shaped parts; Li came up with an element that repeatedly deduces the borders of feature designs. Furthermore, researchers started using Generative Adversarial Networks to obtain a complete image from a part of it. Wang suggested a solution depending on conditional GANs to solve the problem of characteristic extension and framework prediction; Teterwak recommended semantic conditioning for image expanding; Guo implemented image projection.

B. Generative Adversarial Networks

Since 2014, when Goodfellow has created a powerful AI tool, Generative Adversarial Networks have evolved into a considerable concept for image construction. Therefore, many studies show that StyleGANs offer a different architecture that can increase GAN's capacity to generate by borrowing from style transfer literature. The new architecture provides an automatically learned, independently separation of high-level attributes and stochastic variation in the resulted images. GauGAN uses the Gaussian distribution to turn rough doodles into photorealistic images. This way, it offers a powerful tool for creating virtual worlds for architects, urban planners, landscape designers and game developers, prototyping ideas and making fast changes to a virtual scene. The BigGAN has an important contribution to both the models and the training process, creating high-resolution and high-fidelity pictures. These examples prove that GAN has application in various domains, such as fashion, art, science, video games and transfer learning.

C. Generative Image Inpainting with Auxiliary Contextual Reconstruction

Yu Zheng, Zhe Lin, Hunchuan Lu and Vishal M. Patel have proposed a new learnable method to make the generator network to get information from convenient known portions as help for completing the missing part. They also come up with an efficient generator ready to combine the traditional inpainting technique and the auxiliary reconstruction. This solution, also, uses GANs, a machine that generates images and a discriminator. The PatchGAN discriminator has the task to establish the training process. Images that previously have been obtained with inpainting are given to the PatchGAN, which returns a score map designed to maintain a connection between a score and a portion of the input image. The used formula contains matrices that represent the picture, the unknown image with the missing pixels, a simple binary mask that indicates if the pixel is missing or not and the generator. The solution of the operations considers both the unknown portion and the input part. This study gives remarkable results: visual quality, high performance in evaluation, efficiency, capability of processing high-resolution input.

D. Parallel Multi-Resolution Fusion Network

In their study, Wentao Wang, Jianfu Zhang, Li NIu, Haoyu Ling, Xue Yang, and Liqing Zhang have designed a different part-painting architecture, that utilizes multi-resolution partial convolution and masked representation fusion. Partial convolution has importance in eliminating the color conflict problems, using a feature map and a mask. The convolution process is supposed to share details about the known parts to the missing portions. Therefore, a convolution window is obtained and then, updated in order to be passed to the next step: self-attention mechanism. This technique can get global characteristics of the image and obtain a more effective picture generator. Then, the mask-aware representation fusion process begins: the application separates the aspects of the masked and unmasked parts. This step is better than partial convolution because it considers specific cases for pixels (some can include important details, while others can involve information that is

hard to be inpainted). This method obtains practical and realistic results, showing that using fusion techniques create a better connection between the structural and textural component of the images. Moreover, experiments prove that the solution can be applied on regular and irregular input data.

E. Wavelet-based Generation Network

Yu, Zhan, Lu, Pan, Ma, Xie, and Miao present a new generation network, named WaveFill, based on wavelet, that uses discrete transformation, normalization scheme, and LI reconstruction loss to dissolve images into various frequency bands and fill the lacking parts separately. The wavelet transform is a mathematical technique which can decompose a signal into multiple lower resolution levels by conducting the factors of a single wavelet function. Firstly, the application decomposes images into numerous frequency bands, using discrete wavelet transformation (DWT) and completes the parts from the different bands. The next step is to adjust and combine the components that have high and low frequency for generating practical and reasonable content. This important phase is called *Frequency Region Attentive Normalization*. The presented method shows immediate effectiveness qualitatively and quantitatively.

F. Distillation-Guided Image Inpainting

Suin, Purohit and Rajagopalan have introduced an image inpainting method, based on distillation, where they supply uninterrupted supervision over the feature layers in the learning process. The method uses cross and self-distillation solutions, an encoder-decoder that has the objective to complete missing block, attention modules and transfer. The encrypter contains convolution overlays and a filtering block, while the decrypter includes, also, an attention module. The encoder utilizes methods like cross-distillation and self-distillation (the process of transferring knowledge from a large part to a smaller one), adaptive completion-block (despite using mask-based convolution, but both techniques use the same layers for both portions). Attention modules are to intervene in the decoder mechanism: instead of using convolution-based modules, the scientists prefer to utilize a method that calculates pixels correspondence. This approach improves inpainting works, effectively and significantly.

G. Conditional Texture and Structure Dual Generation

Guo, Yang and Huang propose an interesting approach for image inpainting, containing different methods for textural and structural components. Moreover, they implement two modules that have the objective to improve the consistency of the result, to connect the structural and textural information, and to make the final image clearer called *Bi-directional Gated Feature Fusion (Bi-GFF)* and *Contextual Feature Aggregation (CFA)*. The Bi-GFF module is designed to concatenate the texture provided by the decoder and the structure components, while the CFA module is proposed to learn which visible parts of the image can be partly responsible for the missing regions, maintaining a continuous interrelationship between the local and global domain. Compared to other methods, this approach can generate sharp edges and clear textures, elegant details and a strong connection between texture and structure. Experiments made by the scientists prove that this model can give reasonable results.

H. Image-to-Image Translation

Image-to-image translation designs deep neural networks for learning to translate parametrically and functionally a given image. For example, *Pix2Pix* framework uses a conditional adversarial network to learn making a connection between input and output pictures. Both keep the same semantic details, but modifies some information or styles, using a training dataset of cross-domain image pairs. It contains a generator model that takes an image as an input and uses layers that are provided both during the training and when the prediction is made, a discriminator model that predicts likelihood of whether the image from the global domain is real or generated and a loss function that decides if the output images are plausible. The application demonstrates that its methods can be applied on various images, including translating photographs from day to night and products sketches to pictures.

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